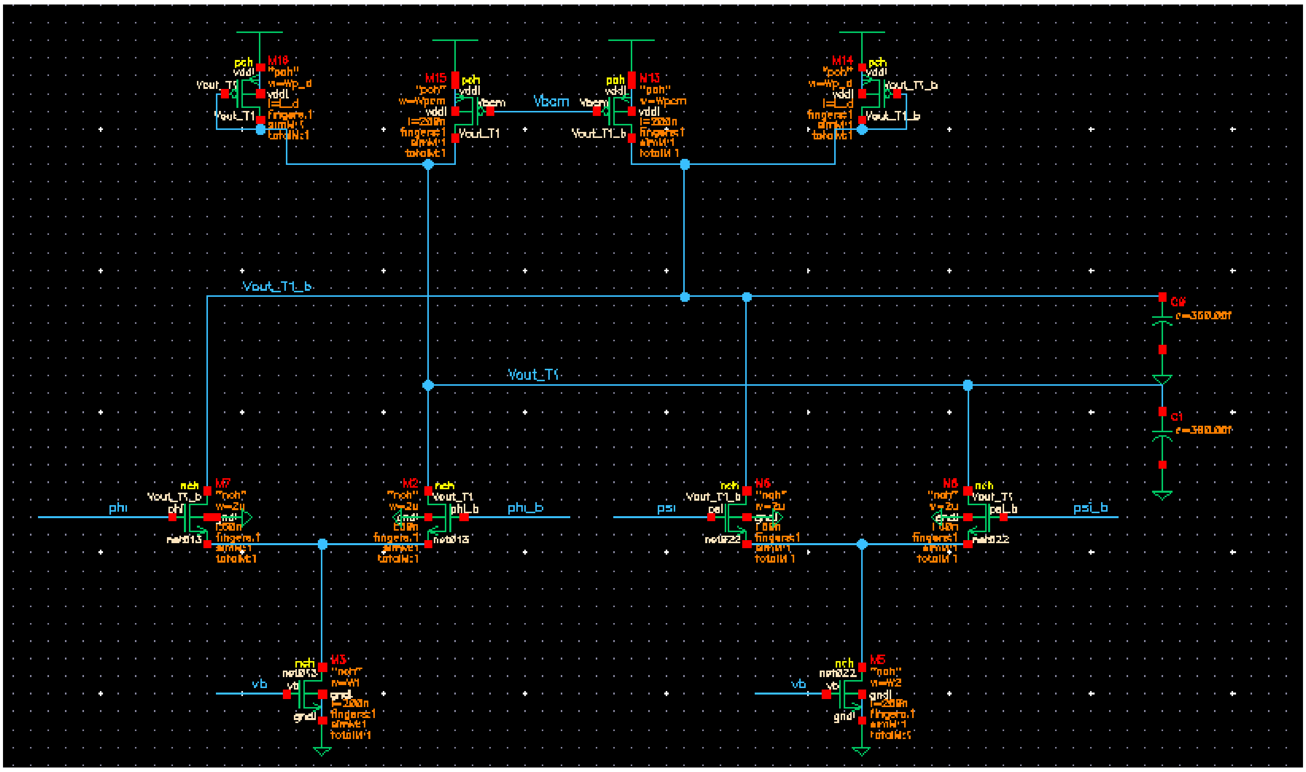
Lab 9
Phase Interpolator

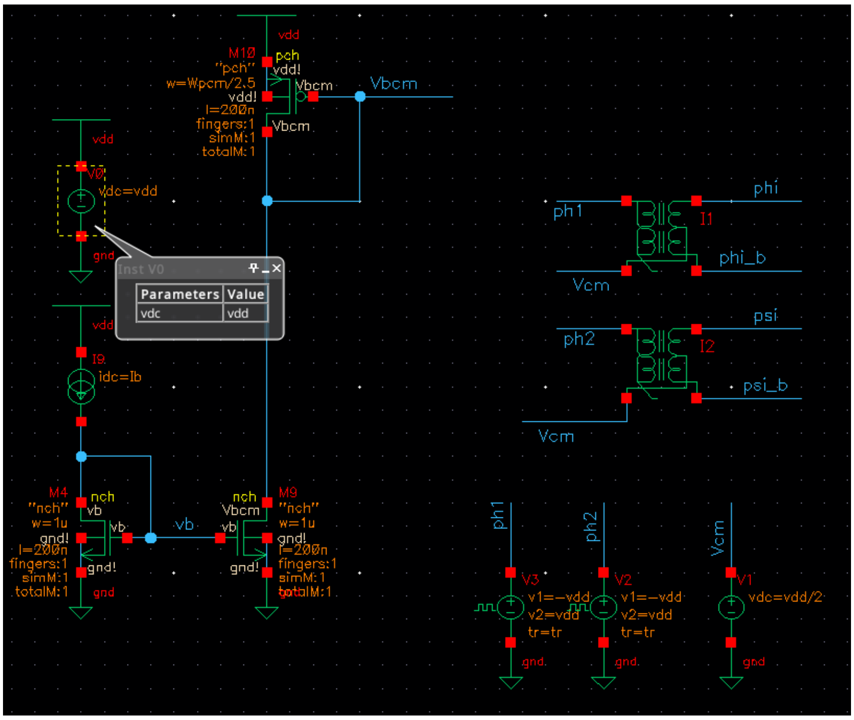
PART 1: Phase Interpolator Circuit

1. Build the following circuit in cadence



The circuit is made of two differential pairs with maneatis load (diode and mirror load back-to-back)
The Sizing of the circuit:

Transistor	W	L
Input Pairs	**2u**	**60n**
Tail Current Source	**5u (make variable W1 and W2)**	**200n**
Pmos Load/Diode	**5u**	**200n**
Reference current mirrors	**1u**	**200n**



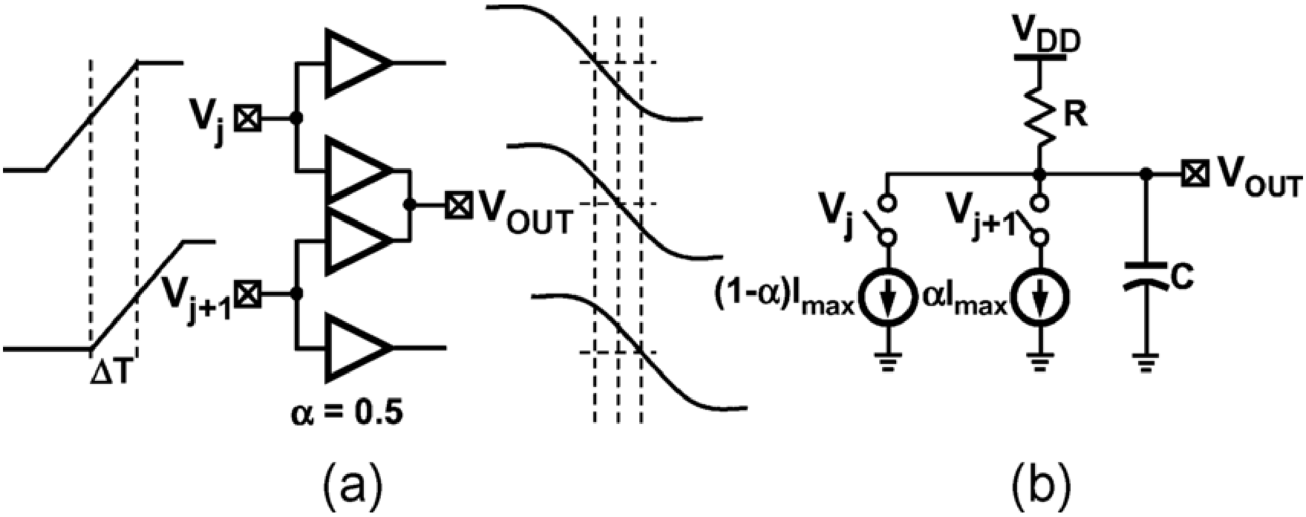
Global Variables	
Ib	20u
vdd	1.2
W1	5u-W2
W2	0.5u*code
Wp_d	Wpcm
Wpcm	5u
vid	500m
Vcm	800m
tr	500p
td	2.5n
L_d	200n
period	20n
code	1:1:9

CDF Parameter	Value	Display
Frequency name for 1/period		off
Noise file name		off
Number of noise/freq pairs	0	off
DC voltage		off
AC magnitude		off
AC phase		off
XF magnitude		off
PAC magnitude		off
PAC phase		off
Voltage 1	-vdd V	off
Voltage 2	vdd V	off
Period	period s	off
Delay time		off
Rise time	tr s	off
Fall time	tr s	off
Pulse width	period/2 s	off

Source phi and other source same with delay time equal to variable td

Inputs (first is $phi\ (\varphi)\ with\ no\ phase\ shift\ the\ second\ input\ is\ Psi\ (\psi)\ with\ phase\ shift = td$

Notes:
The phase interpolator used to get intermediate phases between two inputs separated by 90 degree phase.
To do That we have two differential pairs each phase clock coming from a VCO with multiple phases or DLL go through a 4:2 MUX and 2 phases will be selected having 90 degress phase between.
The current weight from each diff pair will determine phase shift as one is active before the other so the Slewing will change depending on the split ratio
We will sweep over the current weight in tail instead of repeating many cells. and using Variable code.
Simple model shown below where we control value of α with "digital code" represented by variable code.



2- Transient analysis

- **Run Transient Analysis as the following:**

1- Parametric sweep $code = 1:1:9$,$t_r = 500p$ (rise time), $t_d = 2.5n$ (delay between input phases) and $C_{out} = 0.45p$.
Plot $V_{out_{differential}}$ with both input differential in one graph.
Add your comment.

2- Varying RC on delay. same code sweep but compare $V_{out_{differential}}$ when C_{out} = 0.45p and 0.9p, also using the calculator **plot** "phaseDeg($V_{out_{differential}}$)" for both and comment on phase steps.

3- Phase Delay Step

Using function ***delay*** compare the delay of $\boldsymbol{V_{out_{differential}}}$ with the first input signal ($\boldsymbol{\varphi_{differential}}$) then show the phase steps for each value of code. what was the ideal phase step value? calculate the maximum deviation from ideal value → Abs(Max(step - ideal step)/ideal step *100)

*Note: Use the function **Delay** as following

Then **option → falling edge** , **first edge , multiple occurrence** .
The command should be : the subtraction of 3.525n is for normalizing the delays relative to each step. may change with steps edit the value such that first is around 0 degree.

This code will help you to plot delays in degrees if you use the same variables names [you may need to edit it].

```
(delay(?wf1 phi_diff ?value1 0 ?edge1 "falling" ?nth1 1 ?td1 0.0 ?wf2 Vout_diff ?value2 0 ?edge2 "falling" ?nth2 1 ?td2 nil ?stop nil ?period1 1 ?period2 1 ?multiple t ?xName "trigger") - 3.525n)*(360/10n)
```

Comment on the Results.
Is this range of values expected? why?

Then **Plot** delay vs code choose the delay waveform then press axis Tab upper left then choose swap sweep variable.

4- Effect of decreasing relative delay between input phases

Change td = 1n and Cout=0.45 still as above then simulate vout and as number 3 get the delay steps compare then results (range,max deviation from the ideal step) comment.